working memory, *procedural memory*, and *episodic memory*. Much like how humans process events in their working memory [46], FINCON agents leverage their working memory to perform a range of tasks, including observation, distillation, and refinement of available memory events, all tailored to the specific roles of the agents. *Procedural memory* and *episodic memory* are critical for recording historical actions, outcomes, and reflections during sequential decision-making. Procedural memory is generated after each decision step within an episode, storing data as memory events. For trading inquiries, top events are retrieved from procedural memory and ranked based on recency, relevance, and importance, following a simplified version of the method proposed by Yu et al. [32], with further details provided in Appendix A.13. Each functional analyst agent has distinct procedural memory decay rates, reflecting the timeliness of various financial data sources, which is crucial for aligning multi-type data influencing specific time points and supporting informed decision-making. The manager agent enhances the procedural memory of analyst agents by providing feedback through an access counter. Both analyst and manager agents maintain procedural memory, but they keep different records, as illustrated in Appendix A.4. *Episodic memory*, exclusive to the manager agent, stores actions, PnL series from previous episodes, and updated conceptual investment beliefs from the risk control component.

4 Experiments

Our experiment answers the key research questions (RQs): **RQ1:** Does FINCON demonstrate robustness across multiple financial decision-making tasks, especially single-asset trading and portfolio management? **RQ2:** Is the within-episode risk control mechanism in FINCON effective in maintaining superior decision-making performance? **RQ3:** Is the over-episode risk control mechanism in FINCON effective in timely updating the manager agent's beliefs to enhance trading performance?

4.1 Experimental Setup

(i) Multi-Modal Datasets. We construct a market environment representation using real-world financial data, including stock prices, daily news, company filings (Form 10-Q, Form 10-K, etc.), and ECC audio from January 3, 2022, to June 10, 2023, as detailed in Appendix. A.8. Each data source is assigned to specific analyst agents based on its timeliness. ***(ii) Evaluation Metrics.*** We evaluate FINCON and other state-of-the-art (SOTA) agents using metrics such as Cumulative Return (CR%), Sharpe Ratio (SR), and Max Drawdown (MDD%). CR and SR are prioritized because they provide comprehensive insights into overall performance and risk-adjusted returns, essential for informed investment decisions. In contrast, MDD focuses on evaluating the potential for significant losses, making it a secondary consideration in this context. Details are provided in Appendix A.10. ***(iii) Comparative Methods.*** For single-stock trading, we compare FINCON with DRL agents (A2C, PPO, DQN) and LLM-based agents (GENERATIVE AGENT (GA), FINGPT, FINMEM, FINAGENT) as well as the Buy-and-Hold (B & H) strategy. For portfolio management, we compare FINCON with Markowitz MV, FinRL-A2C, and Equal-Weighted ETF strategy, with further details provided in Appendix A.12. The detailed experiment parameter configurations of the above agent systems are articulated in Appendix. A.14. ***(iv) Implementation Details.*** All LLM-based agents use GPT-4-Turbo, with temperature set at 0.3. FINCON is trained from January 3, 2022, to October 4, 2022, and tested from October 5, 2022, to June 10, 2023. DRL agents are trained over the period from January 1, 2018, to October 4, 2022, to ensure that there is sufficient data available for model convergence. Performance is based on the median CR and SR from five repeated epochs. For a more detailed explanation of the experimental setup, please refer to the Appendix A.5.

4.2 Main Results

In response to **RQ1**, we analyze FINCON's performance on two types of financial decision-making tasks: single-asset trading and portfolio management. The system's ability to manage these sequentially complex decisions is thoroughly evaluated in the following sections.

4.2.1 Single Asset Trading Task

In this task, we evaluate FINCON's performance against other leading algorithmic trading models by trading eight different stocks. As presented in the tables above, FINCON significantly outperforms both LLM-based and DRL-based approaches in terms of CRs and SRs. Additionally, FINCON achieves one of the lowest MDD values across most trading assets, demonstrating effective risk management while still delivering the highest investment returns. For detailed performance comparisons across all models and metrics, refer to Table 1.

Overall, even with extended training periods, DRL-based models tend to underperform, with the A2C algorithm lagging significantly behind other agents in general. Notably, the training periods for Nio Inc. (NIO) and Coinbase Global Inc. (COIN) require clarification. NIO, which completed its IPO in September 2018, has a slightly shorter training period than other tickers, yet the DRL algorithms for NIO still achieved convergence. In contrast, Coinbase Global Inc. (COIN), which completed its IPO in April 2021, presented a more significant challenge due to the limited available trading data, causing DRL algorithms to struggle with convergence. This limitation underscores a major drawback for DRL agents when trading recently listed IPOs. Consequently, our analysis of COIN focuses on comparisons between FINCON, LLM-based agents, and the buy-and-hold (B & H) strategy. In this context, FINCON demonstrates a clear advantage, achieving a cumulative return of over 57% and a Sharpe ratio of 0.825. Furthermore, LLM-based agents, which can leverage diverse data types and require minimal training, effectively mitigate the challenges faced by DRL algorithms.

Categories	Models	TSLA			AMZN			NIO			MSFT		
		CR% ↑	SR↑	MDD% ↓	CR % ↑	SR↑	MDD% ↓	CR% ↑	SR↑	MDD% ↓	CR% ↑	SR↑	MDD% ↓
Market	B&H	6.425	0.145	58.150	2.030	0.072	34.241	-77.210	-1.449	63.975	27.856	1.230	15.010
Our Model	FINCON	82.871	1.972	29.727	24.848	0.904	25.889	17.461	0.335	40.647	31.625	1.538	15.010
LLM-based	GA	16.535	0.391	54.131	-5.631	-0.199	37.213	-3.176	-1.574	3.155	-31.821	-1.414	39.808
	FINGPT	1.549	0.044	42.400	-29.811	-1.810	29.671	-4.959	-0.121	37.344	21.535	1.315	16.503
	FINMEM	34.624	1.552	15.674	-18.011	-0.773	36.825	-48.437	-1.180	64.144	-22.036	-1.247	29.435
	FINAGENT	11.960	0.271	55.734	-24.588	-1.493	33.074	0.933	0.051	19.181	-27.534	-1.247	39.544
DRL-based	A2C	-35.644	-0.805	61.502	-12.560	-0.444	37.106	-91.910	-1.728	68.911	21.397	0.962	21.458
	PPO	1.409	0.032	49.740	3.863	0.138	28.085	-72.119	-1.352	62.093	-4.761	-0.214	30.950
	DQN	-1.296	-0.029	58.150	11.171	0.398	31.174	-35.419	-0.662	56.905	27.021	1.216	21.458

Categories	Models	AAPL			GOOG			NFLX			COIN		
		CR% ↑	SR↑	MDD% ↓	CR % ↑	SR↑	MDD% ↓	CR% ↑	SR↑	MDD% ↓	CR% ↑	SR↑	MDD% ↓
Market	B&H	22.315	1.107	20.659	22.420	0.891	21.191	57.338	1.794	20.926	-21.756	-0.311	60.187
Our Model	FINCON	27.352	1.597	15.266	25.077	1.052	17.530	69.239	2.370	20.792	57.045	0.825	42.679
LLM-based	GA	5.694	0.372	14.161	-1.515	-0.192	8.210	41.770	1.485	20.926	19.271	0.277	67.532
	FINGPT	20.321	1.161	16.759	0.242	0.011	26.984	11.925	0.472	20.201	-99.553	-1.807	74.967
	FINMEM	12.397	0.994	11.268	0.311	0.018	21.503	-10.306	-0.478	27.692	0.811	0.017	50.390
	FINAGENT	20.757	1.041	19.896	-7.440	-1.024	10.360	61.303	1.960	20.926	-5.971	-0.106	56.882
DRL-based	A2C	13.781	0.683	14.226	8.562	0.340	21.191	-8.176	-0.258	49.579	-	-	-
	PPO	14.041	0.704	22.785	2.434	0.097	25.202	-33.144	-1.049	33.377	-	-	-
	DQN	21.125	1.048	16.131	20.690	0.822	21.191	21.753	0.687	39.733	-	-	-

Table 2: Comparison of key performance metrics during the testing period for the single-asset trading tasks involving eight stocks, between FINCON and other algorithmic agents. *Note that the highest and second highest CRs and SRs have been tested and found statistically significant using the Wilcoxon signed-rank test. The highest CRs and SRs are highlighted in red, while the second highest are marked in blue.*

In alignment with market trends, FINCON consistently exhibits superior decision-making quality compared to other LLM-based agents, regardless of market conditions—whether bullish (e.g., GOOG, MSFT), bearish (e.g., NIO), or mixed (e.g., TSLA). We attribute this performance to its high-quality distillation of information through a synthesized multi-agent collaboration mechanism, combined with its dual-level risk control design, positioning FINCON as a leader in the space. By contrast, FINGPT primarily relies on sentiment analysis of financial information, failing to fully exploit the potential of LLMs to integrate nuanced textual insights with numerical financial indicators. Similarly, GA and FINMEM use single-agent frameworks without sophisticated information distillation processes or a diverse toolset, placing heavy cognitive demand on the agent to process multi-source information, especially when dealing with large and varied data modalities. Moreover, their static or minimal investment belief systems result in weak filtering of market noise. As illustrated in Figure 7 (a) & (b) of Appendix A.7.2, this limitation leads these models to consistently hold lower positions and hesitate between 'buy' or 'sell' decisions, ultimately resulting in suboptimal performance.

FINCON overcomes these challenges through its innovative multi-agent synthesis, enabling it to deliver superior outcomes. Although FINAGENT performs well when integrating images and tabular data, it struggles to remain competitive when incorporating audio data, such as ECC recordings, which are critical in real-world trading. Additionally, FINAGENT relies on similarity-based memory retrieval, which can lead to decisions based on outdated information, often resulting in errors. In contrast, FINCON's memory structure accounts for the varying timeliness of multi-source financial data, significantly enhancing decision quality and overall performance.

4.2.2 Portfolio Management Task

In this task, we compare FINCON's performance with the Markowitz Mean-Variance (MV) portfolio [47] and FINRL [48] in managing two small portfolios: Portfolio 1 (TSLA, MSFT, and PFE) and Portfolio 2 (AMZN, GM, and LLY). These assets were selected by the stock selection agent from a pool of 42 stocks, each with sufficient news data (over 800 news articles during the combined training and testing periods), as illustrated in Figure 9 in Appendix A.9. The training and testing periods, the backbone model and the parameter settings are consistent with those used in the single-asset trading task. For the Markowitz MV portfolio, we estimate the covariance matrix and expected returns using the same training data. In the case of FINRL, we use five years of training data prior to the test period. As detailed in Table 3 and Figure 3, our results show that FINCON outperforms both the Markowitz MV portfolio and FINRL as well as the market baseline – Equal-Weighted ETF, achieving significantly higher CRs and SRs, as well as MDDs.

However, managing multi-asset portfolios introduces more complexity, leading to a higher likelihood of hallucination compared to single-asset trading. This is due to the increased input length and complexity involved in multi-asset decision-making. While FINCON mitigates this issue by distributing tasks across specialized agents that focus on critical investment insights, it occasionally generates incorrect information, such as non-existent indices of memory events. Handling multi-asset decision-making requires sophisticated logic and substantial market information, which poses a significant challenge for LLMs when processing extended contexts. This complexity has left portfolio management relatively unexplored in previous language agent studies. Nonetheless, FINCON demonstrates considerable potential by constructing agent systems that can tackle complex financial tasks through effective resource optimization, even when managing relatively compact portfolios.

Models	CR % ↑	SR↑	MDD %↓
FINCON	113.836	3.269	16.163
Markowitz MV	12.636	0.614	17.842
FINRL-A2C	19.461	0.831	26.917
Equal-Weighted ETF	9.344	0.492	21.223

Models	CR % ↑	SR↑	MDD %↓
FINCON	32.922	1.371	21.502
Markowitz MV	10.289	0.540	25.099
FINRL-A2C	11.589	0.649	15.787
Equal-Weighted ETF	15.061	0.867	14.662

Table 3: Key performance metrics comparison among all portfolio management strategies of Portfolio 1 & 2. FINCON leads all performance metrics.

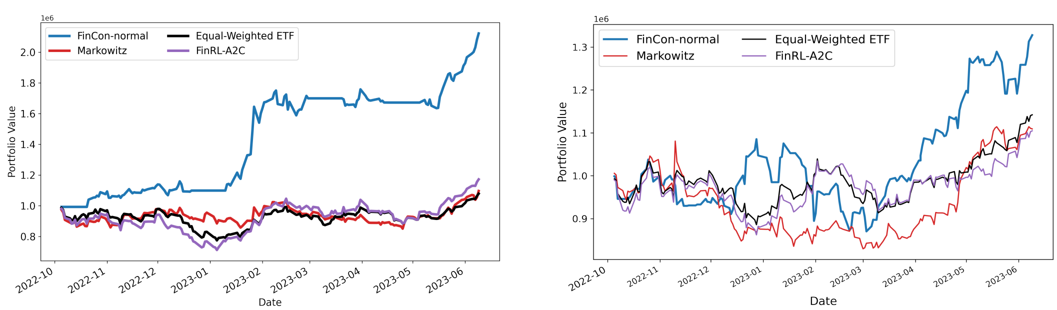

Figure 3: Portfolio values of Portfolio 1 & 2 changes over time for all the strategies. The computation of portfolio value refers to Equation 7 in Appendix A.10.

4.3 Ablation Studies

In response to **RQ2** and **RQ3**, we conduct a comprehensive evaluation of our unique risk control component through two ablation studies. Both studies maintain consistency with the training and testing periods used in the main experiments. The first study examines the effectiveness of the within-episode risk control mechanism, which leverages Conditional Value at Risk (CVaR) to manage risk in real-time, as detailed in Table 4. Comparisons on primary metrics illustrate that the success of utilizing CVaR for within-episode risk control is evident in both bullish and bearish market environments in the